

Laplace transforms of standard functions

1. Find $L\left\{\frac{1}{\sqrt{t}}\right\}$

ans: $\sqrt{\frac{\pi}{s}}$

2. Find $L\{\sin(t)\sin(5t)\}$

ans: $\frac{10s}{(s^2+16)(s^2+36)}$

3. Find $L\{\cosh(at)\}$

ans: $\frac{s}{s^2-a^2} (s > a)$

4. Find $L\{e^{2t} + 3e^{-t} + 5\}$

ans: $\frac{9s^2-10s-10}{s(s+1)(s-2)}$

5. Find $L\{\cos^2(2t-1)\}$

ans: $\frac{1}{2s} + \frac{1}{2} \left(\frac{s \cos(2) + 4 \sin(2)}{s^2+16} \right)$